DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

SMA 3244::: PDE CAT ONE TIME: 1 HOUR

x^2+y^2
$=(x+y)(x-y)$

a) Distinguish between a partial differential equation and ordinary differential equation.

b) Find the integral surfaces of the equation $x(y^2+Z)p-y(x^2+z)q=(x^2-y^2)Z$

c) Solve the given equations; $\frac{dx}{x^2(y^3-z^3)}=\frac{dy}{y^2(z^3-x^3)}=\frac{dz}{z^2(x^3-y^3)}$

d) Find the orthogonal trajectories on the surface $xz+yz=9$ of the conics in which its cut by the system of planes $x-y+z=c$ where c is a parameter.

x^2+y^2

e) Form a Quasi-linear equation whose solution is given by $\phi(x^2+y^2+z^2, xyz)=0$

$z^2-2cz+c^2=a^2$

f) Determine the partial differential equation arising from $x^2+y^2+(Z-c)^2=a^2$